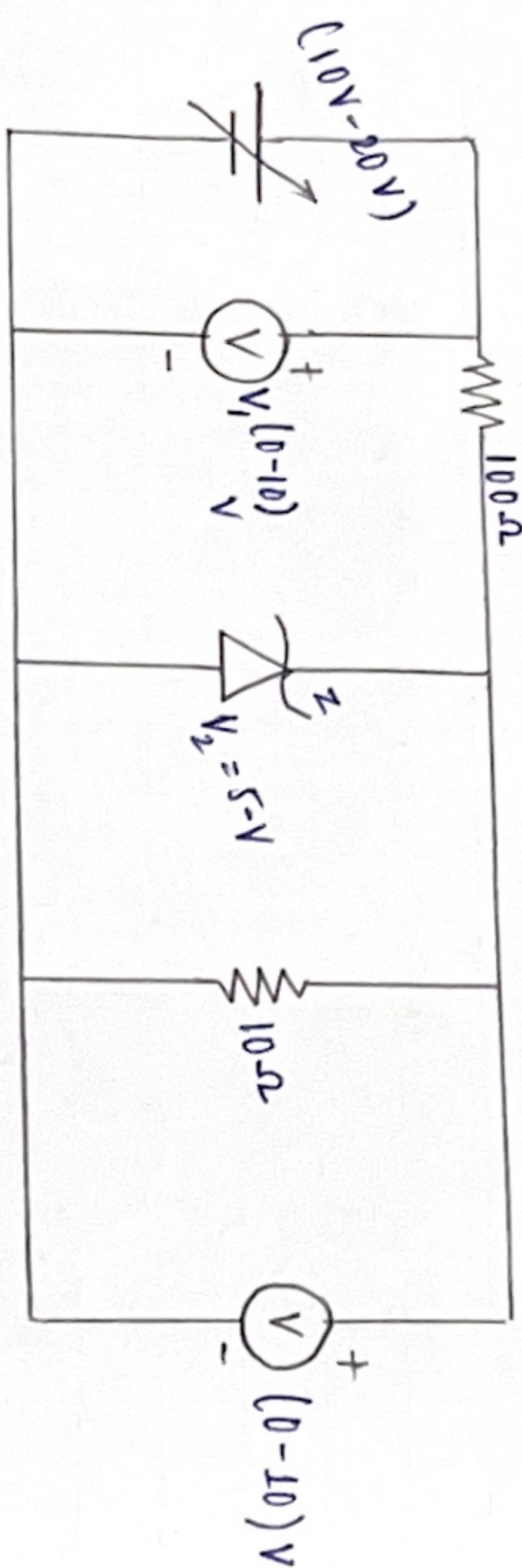


CIRCUIT DIAGRAM:

LINE REGULATION



TABULATION:

S. NO	INPUT VOLTAGE V_1 (V)	OUTPUT VOLTAGE V_2 (V)
1.	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

MODEL GRAPH:



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ZENER DIODE AS VOLTAGE REGULATION

Aim:

To simulate the line and load regulation operation Zener diode.

Apparatus Required:

Laptop with proteus software.

Theory:

A heavily doped PN Junction diode is called Zener diode. Due to heavily doped nature Zener diode works under forward bias and reverse bias conditions.

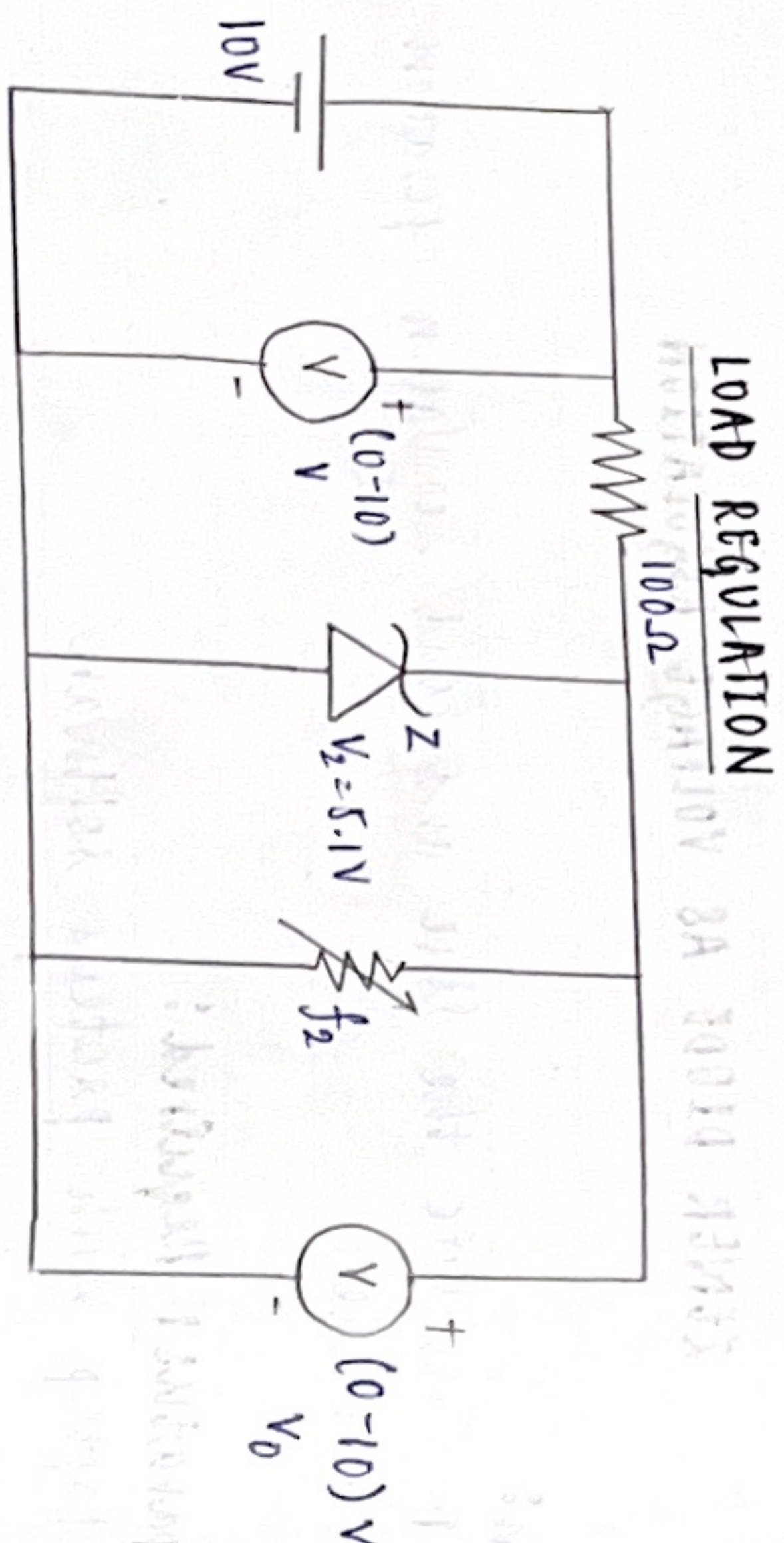
Zener diode under reverse bias condition is used as a voltage regulation. Even if the input voltage changes the output is constant. This is called line regulation.

If the load resistance changes the output voltage is constant this is called load regulation.

Procedure:

1. Drag the required components from the proteus library.
2. Connect the components (required) as per circuit diagram.
3. For line regulation vary the input voltage and note the output voltage in tabulation. Draw the graph between input voltage and output voltage.

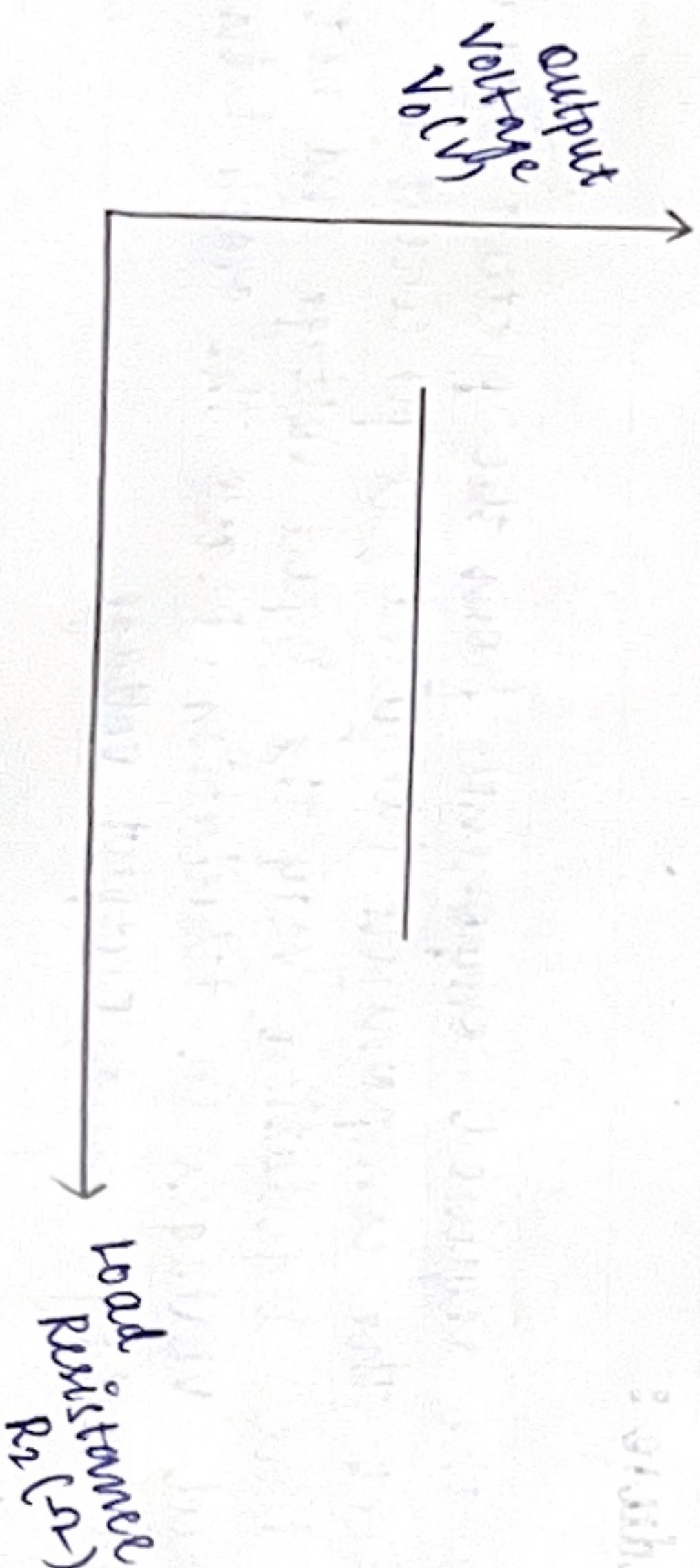
CIRCUIT DIAGRAM:



TABULATION:

S.NO	LOAD RESISTANCE $R_2(\Omega)$	OUTPUT VOLTAGE $V_0(V)$
1	100 Ω	+5.19
2	200 Ω	+5.20
3	300 Ω	+5.20
4	400 Ω	+5.20
5	500 Ω	+5.20
6	600 Ω	+5.21

MODEL DIAGRAM:



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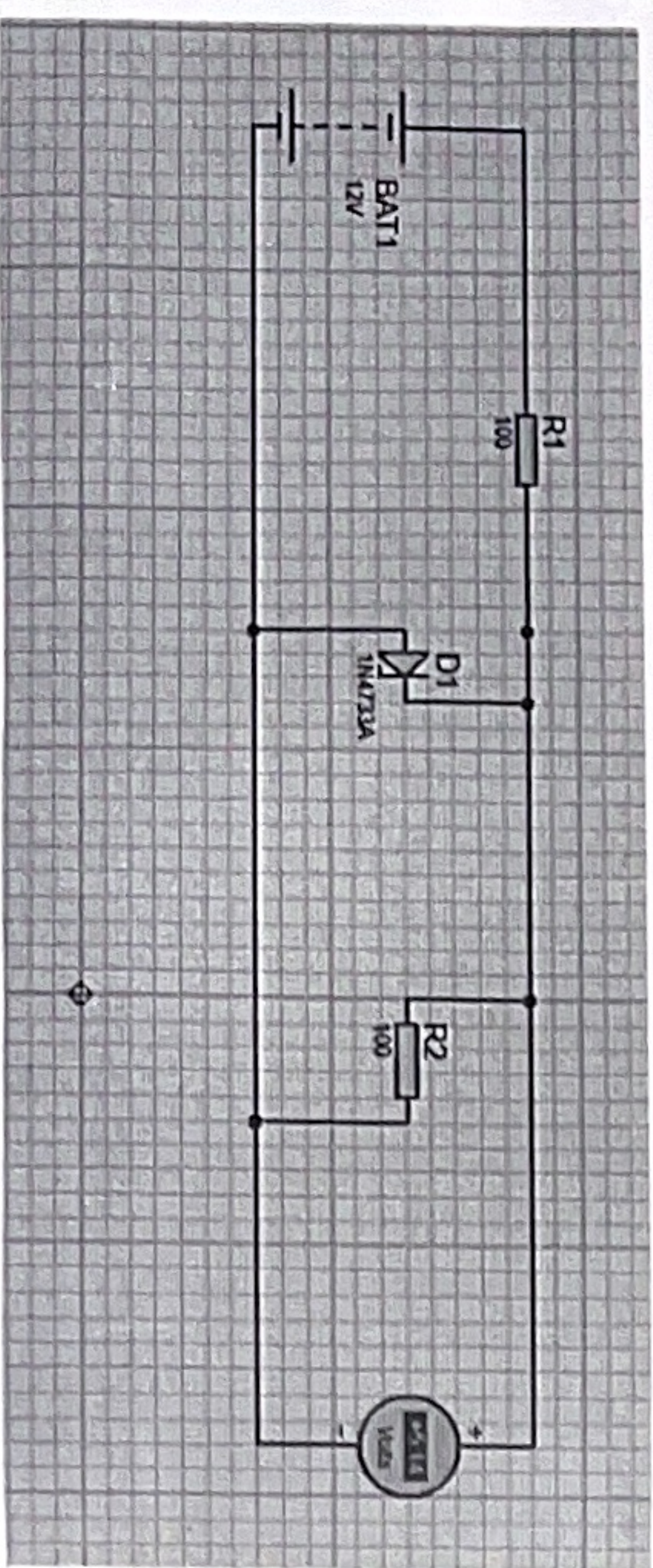
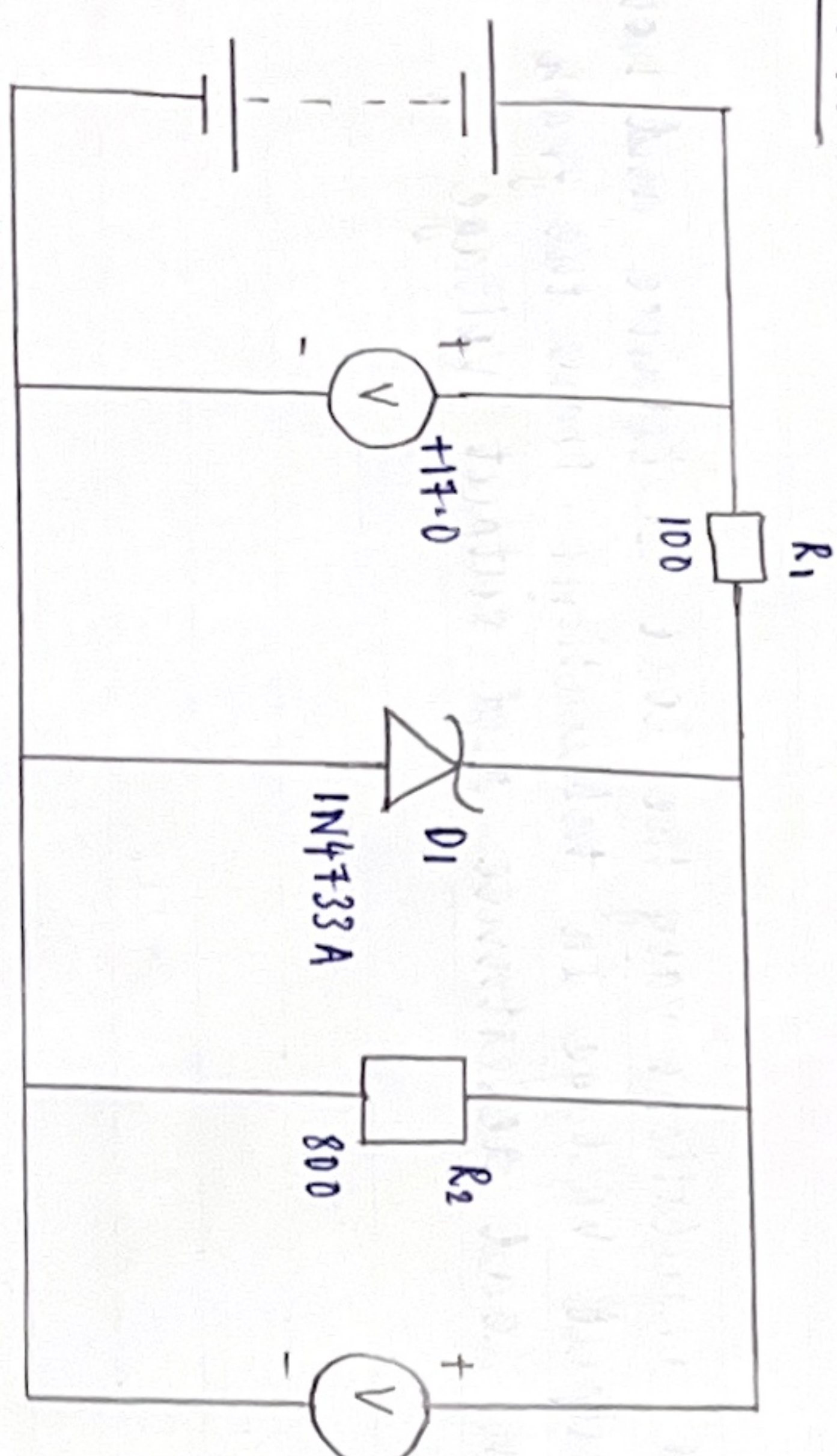
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4.8 For load regulation vary the load resistance and note the output voltage in tabulations. Draw the graph between load resistance and output voltage.



OUTPUT:



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Result:

Thus stimulation of line load regulation operation of Zener diode successfully completed.